# The Reel Deal: Movie Ticket Booking System

System Requirements Documentation

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# Customer Problem Statements & System Requirements

**The Reel Deal: Movie Ticket Booking System**

## Customer Problem Statement:

When I go to a movie theater, I usually have no idea if there are still good seats left for the showtime I want. Even if I get there early, I may still end up waiting in line, only to find that the remaining seats are scattered or already taken. If I go with friends or family, it can be frustrating trying to find seats together. This often wastes time and takes away from the experience before the movie even starts.

Theater staff deal with similar problems. They have to manage movie listings, set up showtimes, keep track of which seats are available, and handle ticket sales all at once. Without a clear system to organize this information, it can lead to delays, confusion, and seat assignment mistakes.

As a customer, I would expect a system where I can see movies and showtimes ahead of time, check which seats are open, reserve seats before arriving, and receive confirmation of my booking. Theater staff would expect a system that helps them manage movies, showtimes, and seating in a simple and organized way.

## Glossary of Terms:

| Term | Definition |
|---|---|
| Movie | A film available for customers to book in the system |
| Showtime | A scheduled time when a movie plays in a specific auditorium |
| Auditorium | A theater room with a defined seating layout |
| Seat | A selectable position in an auditorium |
| Booking | A confirmed reservation of one or more seats for a showtime |
| Ticket | Proof of a completed booking |
| Manager | User role that manages movies, showtimes, and seating |
| Staff | User role that validates tickets and assists customers |

# System Requirements

## Functional Requirements:

| REQ # | Priority Weight | Description |
|---|---|---|
| REQ-1 | High | The system should allow managers to add, edit, and delete movies |
| REQ-2 | High | The system should allow managers to create and manage showtimes |
| REQ-3 | High | The system should allow managers to define seating layouts for auditoriums |
| REQ-4 | High | The system should display seat availability for each showtime |
| REQ-5 | High | Customers should be able to select and reserve available seats |
| REQ-6 | High | The system should prevent double booking of the same seat |
| REQ-7 | Medium | Customers should be able to view their booking history |
| REQ-8 | Medium | The system should support different seat types such as standard and accessible |
| REQ-9 | High | The system should update seat status immediately after booking |
| REQ-10 | Medium | The system should allow users to create accounts and log in |
| REQ-11 | Low | The system should display movie details and descriptions |
| REQ-12 | High | Theater staff should be able to validate tickets for customers |

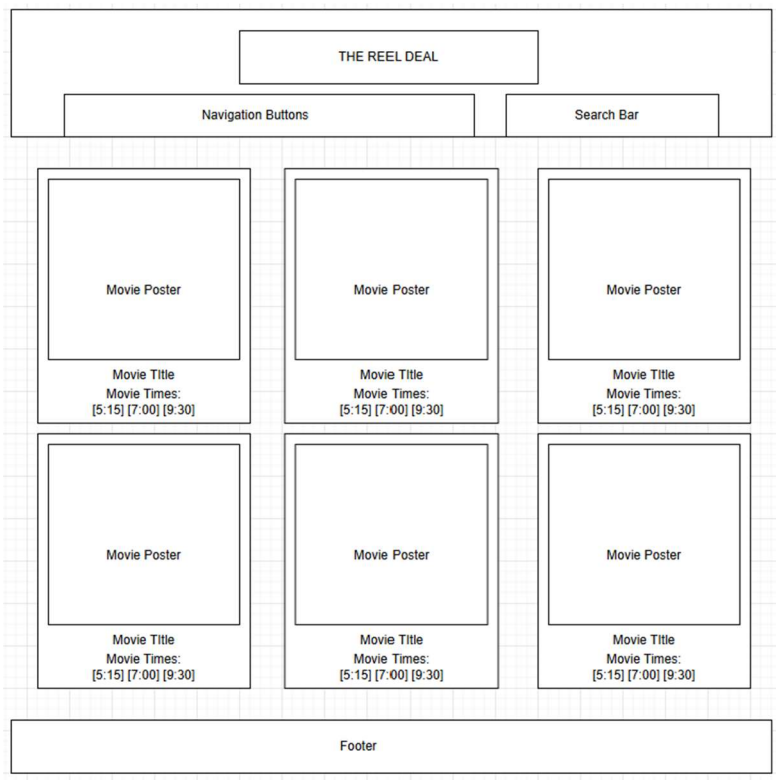
## Nonfunctional Requirements (FURPS):

| NFR # | Priority Weight | Description |
|---|---|---|
| NFR-1 | High | Seat availability should update within 2 seconds after a booking (Performance) |
| NFR-2 | High | The system should prevent data loss during booking transactions (Reliability) |
| NFR-3 | Medium | The system should be easy for first-time users to navigate (Usability) |
| NFR-4 | Medium | The system should be accessible from standard web browsers (Supportability) |
| NFR-5 | Medium | The interface should clearly show seats and availability (User Interface) |

## User Interface Requirements:

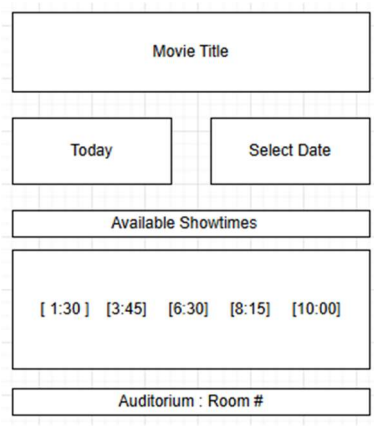
The following user interface requirements describe the primary screens that users interact with when using the movie booking system.

UI-1: Movie List Page
The system should display all available movies and their showtimes.



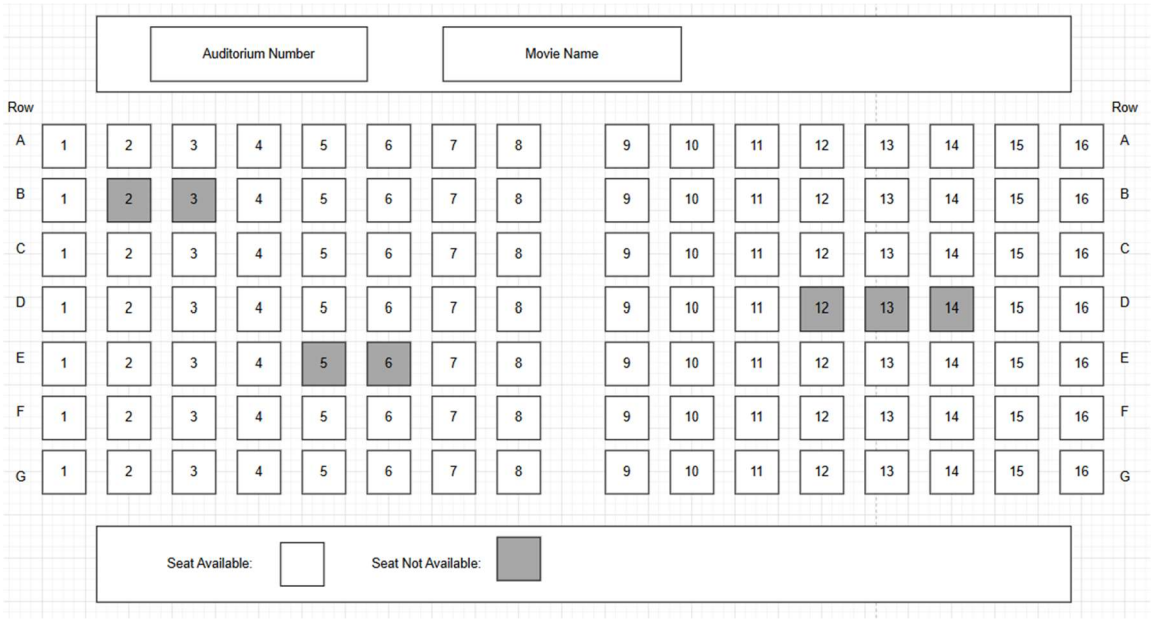
UI-2: Showtime Selection Page

The system should display available showtimes for a selected movie.



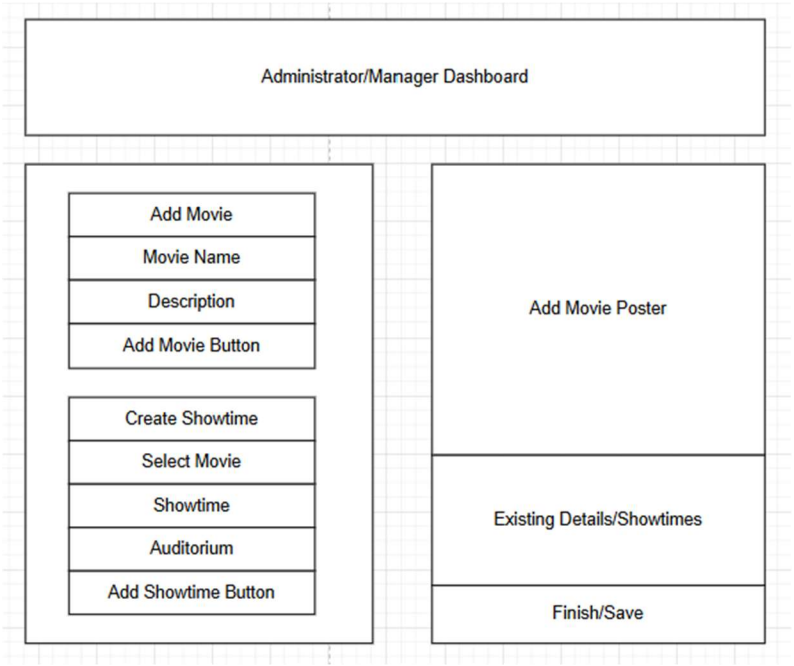

UI-3: Seat Selection Page

The system should display a seating layout with available and reserved seats clearly marked.



UI-4: Manager Control Page

The system should allow managers to add movies, create showtimes, and manage seating layouts.




UI-5: Booking Confirmation Page

The system should display booking confirmation details after a successful seat reservation.

# Functional Requirement Specification

**Stakeholders**

- Theater owner who wants an organized and efficient ticket booking system
- Theater manager responsible for managing movies, showtimes, and seating layouts
- Theater staff who validate tickets and assist customers
- Customers who want to reserve seats and purchase tickets in advance
- System developer responsible for building and maintaining the system

**Primary Actors**

- Customer: This actor can browse movies, select showtimes, view available seats, reserve seats, and receive a ticket confirmation.
- Theater Staff: This actor can validate customer tickets and view booking information to assist customers.
- Theater Manager: This actor can add, edit, and remove movies, create showtimes, and manage seating layouts.

**Secondary Actors**

- System: This actor is responsible for updating seat availability, storing booking records, and preventing double booking of seats.

## Use Cases

**Theater Manager (total: 16)**

- Add movie: To add a new movie to the system (3)
- Edit movie: To update movie details (2)
- Create showtime: To schedule a new showtime for a movie (3)
- Edit showtime: To update or remove a showtime (2)
- Define seating layout: To create and manage the seating layout of an auditorium (3)
- Login/Logout: To log in and out of the manager account (3)

**Customer (total: 14)**

- View movies: To see available movies and showtimes (2)
- Select showtime: To choose a showtime for a movie (2)
- View seats: To see which seats are available or reserved (3)

- Reserve seat: To select and reserve available seats (3)
- Confirm booking: To finalize the booking and receive ticket confirmation (2)
- View booking history: To view previous bookings (2)

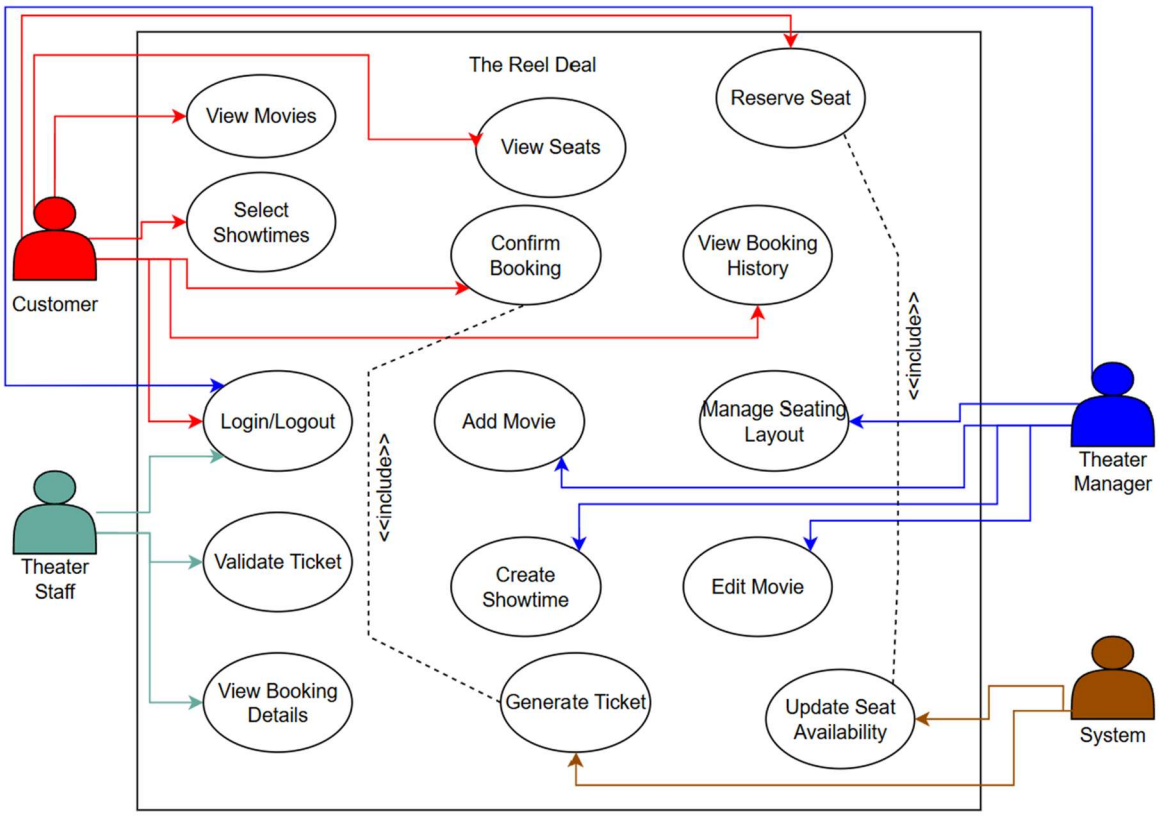
**Theater Staff (total: 6)**

- Login/Logout: To log in and out of the staff account (2)
- Validate ticket: To confirm the validity of a customer's ticket (2)
- View booking information: To check booking details when assisting customers (2)

**System (total: 8)**

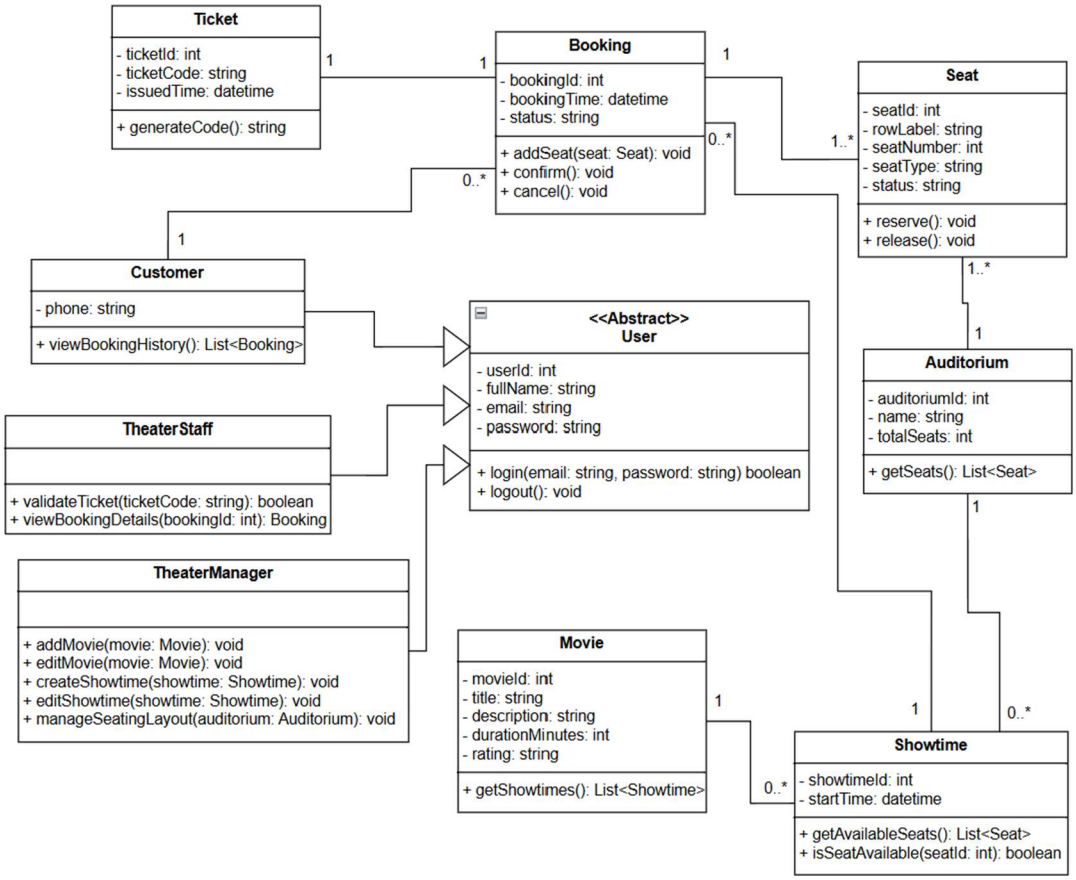
- Update seat availability: To automatically update seat status after booking (3)
- Prevent double booking: To ensure a seat cannot be reserved by more than one customer (3)
- Store booking records: To save booking and ticket data in the database (2)

## Use Case Diagram



## Class Diagram

The class diagram models movies, showtimes, auditoriums, seats, bookings, and tickets. Users inherit from an abstract User class to represent customer, staff, and manager roles. Multiplicity shows how one movie can have many showtimes, and one booking can include multiple seats. This diagram also enforces the rule that a seat can only be reserved once per showtime through the relationship between Booking, Seat, and Showtime.



# System Sequence Diagram and Activity Diagram

## System Sequence Diagrams:

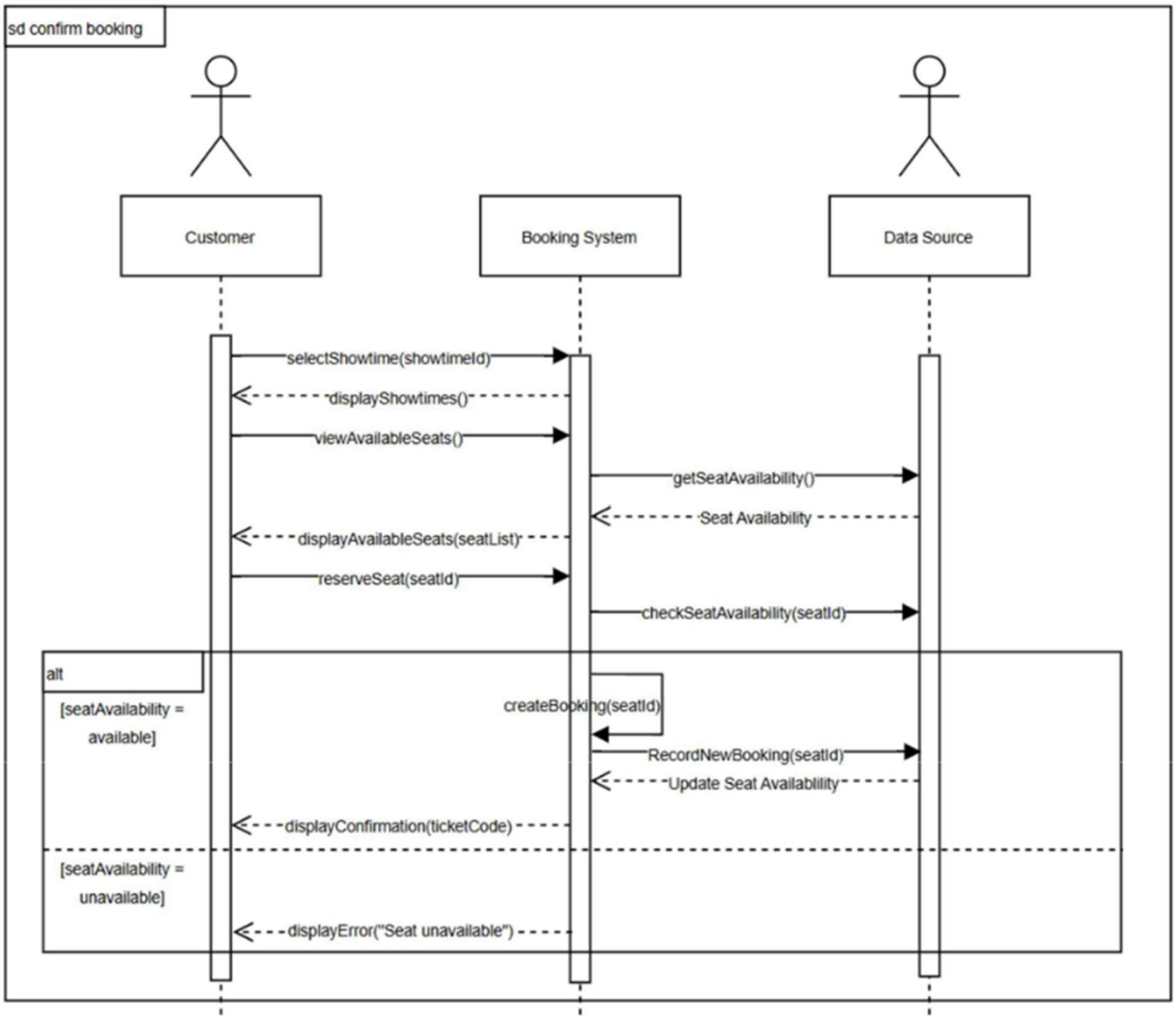
### Sequence Diagram for Confirm Booking

Actor: Customer
Object: Booking System
Interacts with the Data Source or Database

Steps of confirming a booking:

1. The Customer selects a movie and showtime.
2. The Customer requests to view available seats for the selected showtime.
3. The Booking System retrieves seat availability information from the data source or database.
4. The Booking System displays available seats to the Customer.
5. The Customer selects one or more seats and requests to reserve them.
6. The Booking System checks seat availability in the data source or database.
7. If the seats are available:
    1) A new booking record is created in the data source or database.
    2) The selected seats are marked as reserved in the data source or database.
    3) A ticket is generated and stored in the data source or database.
    4) A booking confirmation screen with ticket details is shown to the Customer.
8. If one or more seats are not available:
    1) The Booking System displays an error message indicating the seat is unavailable.
    2) The Customer is prompted to select different seats.

The sequence diagram is shown below:



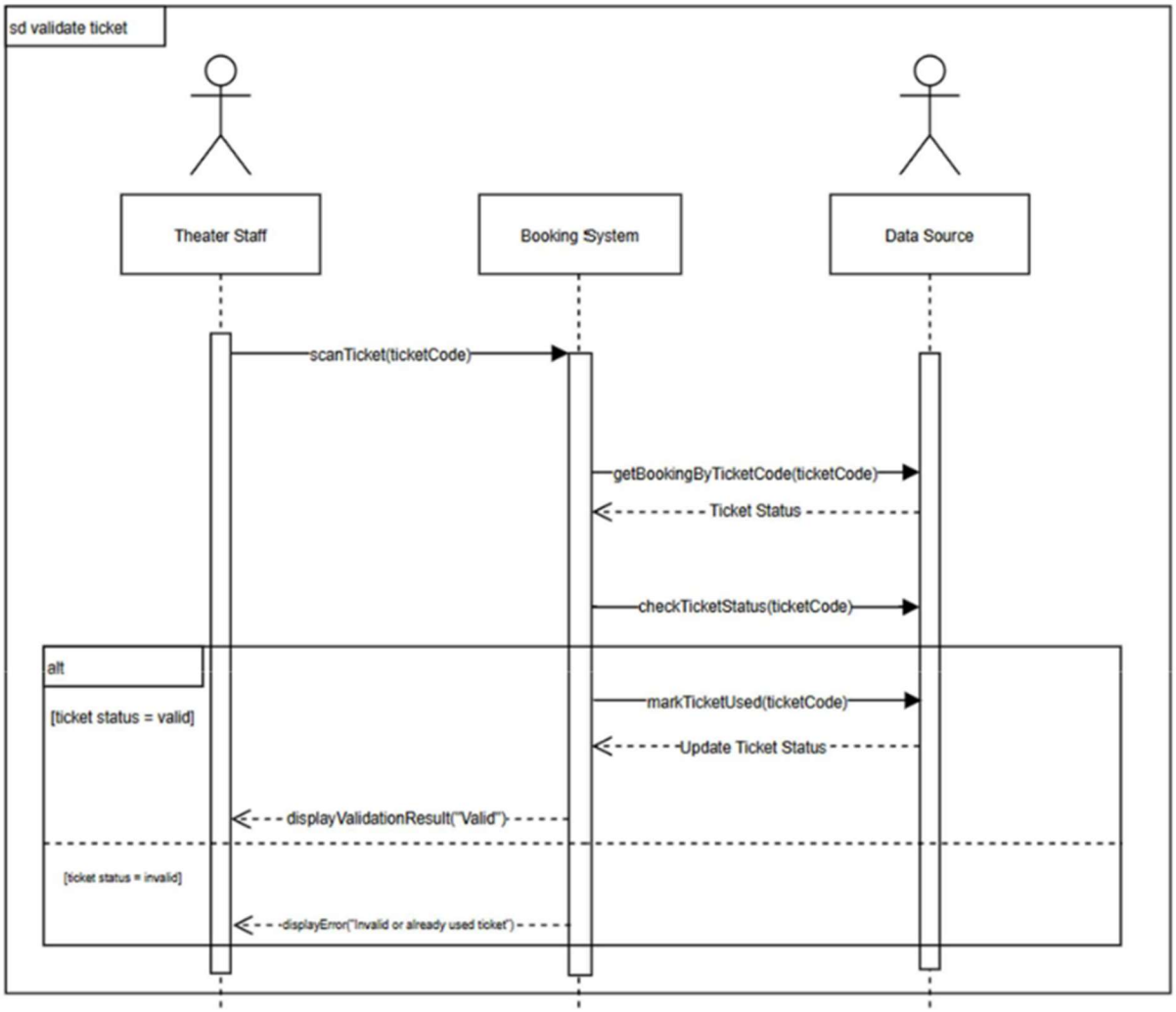
## Sequence Diagram for Validate Ticket

Actor: Theater Staff
Object: Booking System
Interacts with the Data Source or Database

Steps of validating a ticket:

1. The Theater Staff scans or enters the ticket code.
2. The Theater Staff submits the ticket code to the Booking System for validation.
3. The Booking System retrieves the booking record associated with the ticket code from the data source or database.
4. The Booking System checks the ticket and booking status (for example: exists, not expired, not already used).
5. If the ticket is valid:
    1) The Booking System updates the ticket status to “used” in the data source or database.
    2) The Booking System displays a “Valid Ticket” result to the Theater Staff (entry allowed).
6. If the ticket is invalid (not found, already used, or canceled):
    1) The Booking System displays an error message to the Theater Staff (entry denied).

The sequence diagram is shown below:



# Activity Diagrams:

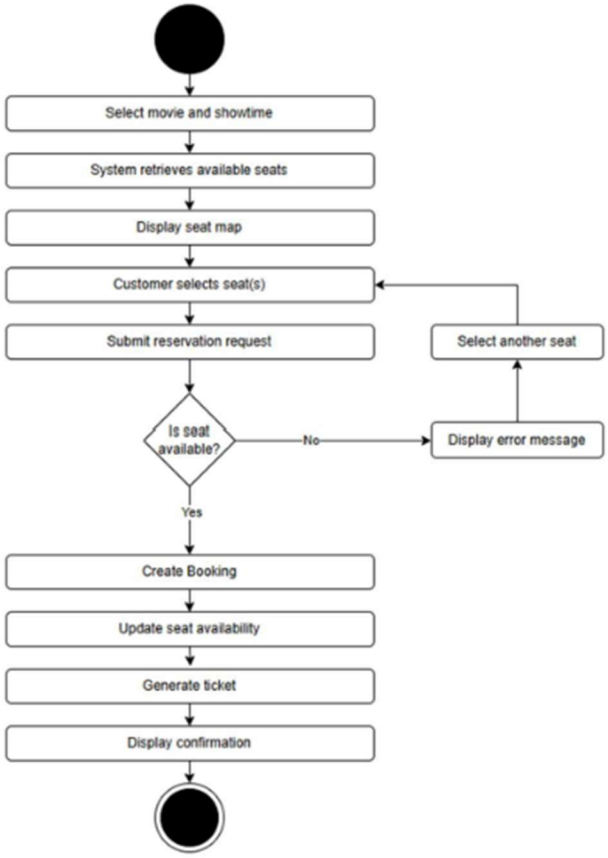
## Confirm Booking

**States**

- Initial State: The customer selects a movie and showtime to book tickets.
- Final State:
    1. The customer receives a booking confirmation with a ticket code.
    2. The selected seat is unavailable, and the customer receives an error message and must choose another seat.

**Actions**

The customer selects a showtime and requests to view available seats. The system retrieves seat availability and displays the seat map. The customer selects a seat and submits a reservation request. The system checks whether the seat is still available. If it is available, the system creates a booking, updates seat availability, generates a ticket code, and displays the confirmation. If the seat is not available, the system displays an error message and prompts the customer to select a different seat.



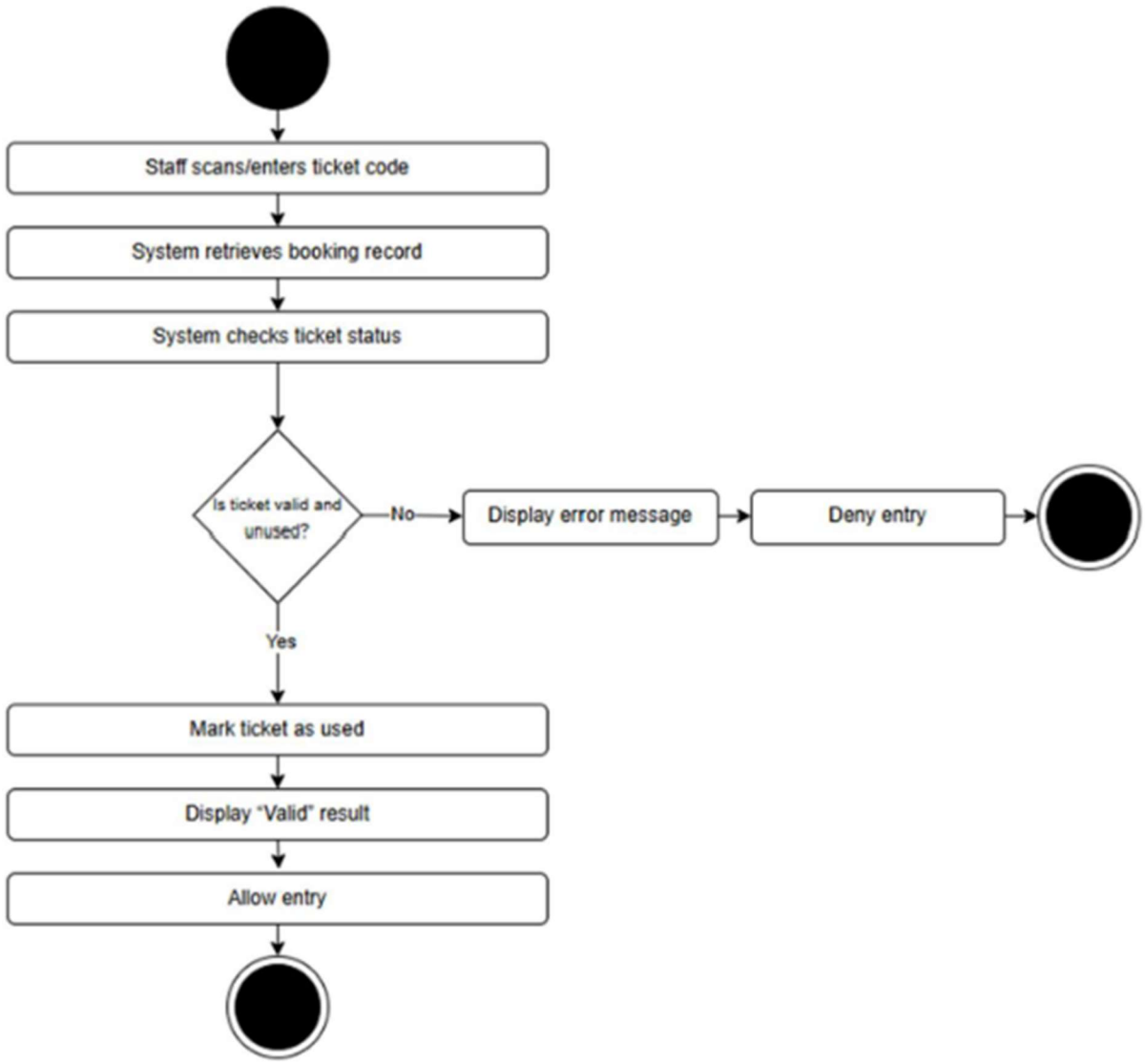
## Validate Ticket

**States**

- Initial State: Theater staff scans or enters a ticket code to validate entry.
- Final State:
    1. The ticket is valid, and the system approves entry.
    2. The ticket is invalid (not found, canceled, or already used), and the system denies entry.

**Actions**

Theater staff scans or enters a ticket code at the theater entrance. The system retrieves the booking information associated with the ticket code and checks the ticket status. If the ticket is valid and unused, the system marks the ticket as used and displays a valid result to staff. If the ticket is invalid or already used, the system displays an error message and entry is denied.



# User Interface Specification

This document outlines the initial user interface design for The Reel Deal movie booking system. It shows how users will interact with the system and what each screen will display. The mockups are included to give a clear idea of how users navigate the system, enter information, and receive feedback after completing actions like selecting seats or confirming a booking.

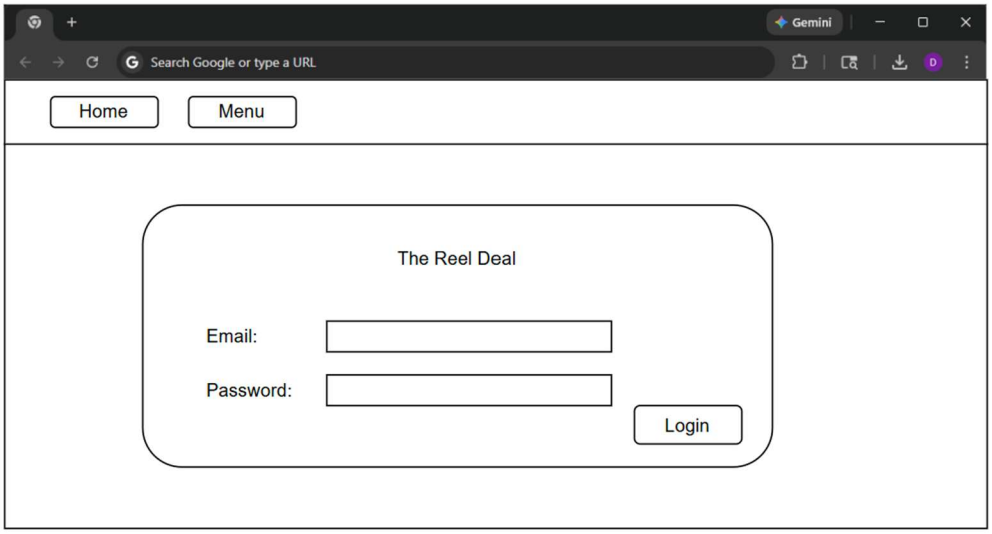
The following use cases are included:

- Login
- View Movies and Showtimes
- Select Seats and Confirm Booking
- Display Booking Confirmation
- Validate Ticket

## Use Case 1: Login

**Preliminary Design:**
The login screen lets users sign in by entering their email and password. There are two input fields and a Login button. Once the user enters their information and clicks Login, the system verifies the credentials and moves them to the movie selection screen if the login is successful. If the information is wrong, an error message is displayed so the user can correct it.

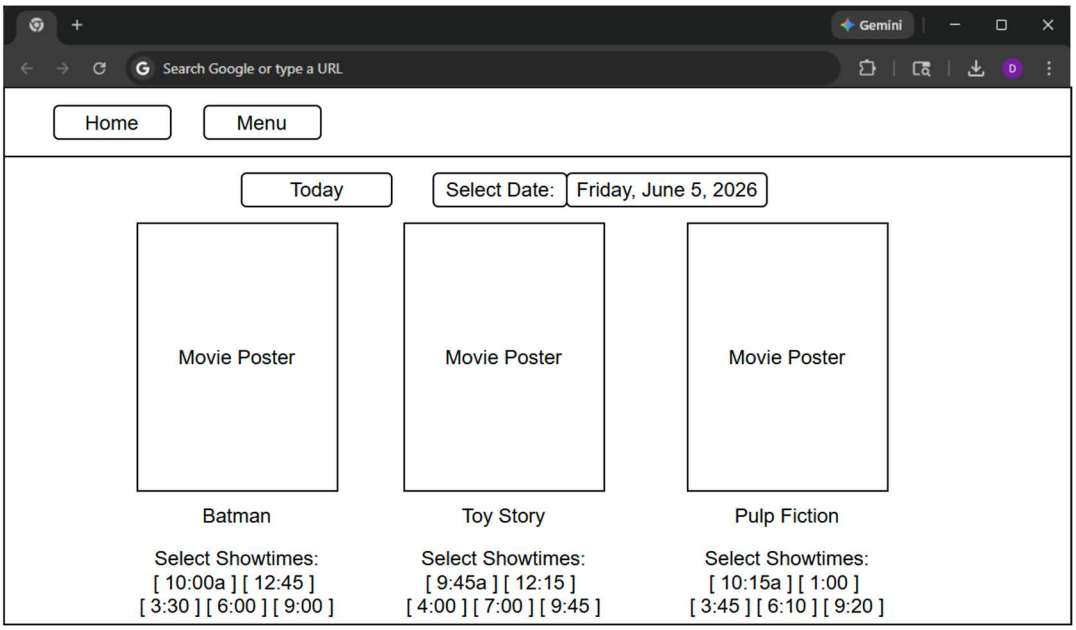
**Navigation Path:**
Home Screen > Login Screen > Movie Selection Screen



## Use Case 2: View Movies and Select Showtime

**Preliminary Design:**
The movie selection screen lets users browse the movies that are currently available, along with their posters, titles, and showtimes. Each showtime is displayed as a clickable button. Once the user clicks a showtime, the system moves them to the seat selection screen for that movie so they can continue the booking process.

**Navigation Path:**
Login Screen > Movie Selection Screen > Seat Selection Screen

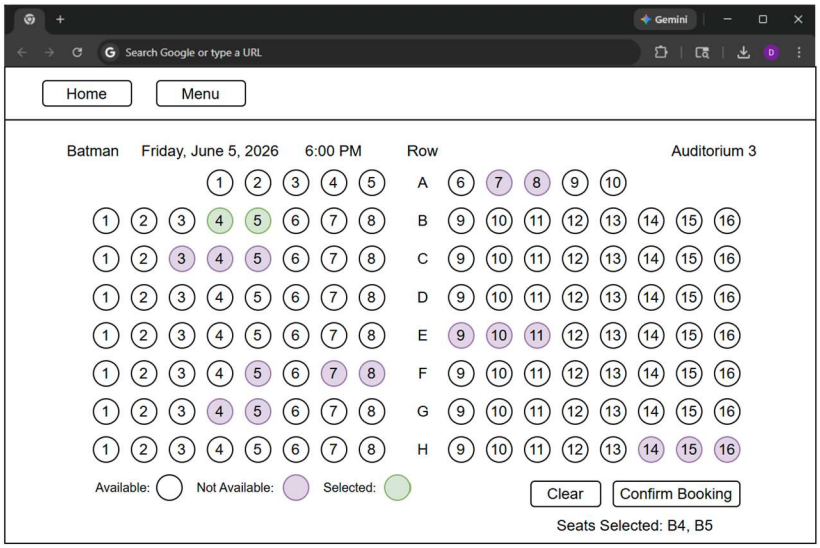



## Use Case 3: Select Seats and Confirm Booking

**Preliminary Design:**
The seat selection screen lets the user see the seating layout for the selected movie and showtime. It clearly shows which seats are available and which ones are already taken. The user can click on available seats to select them and then click the Confirm Booking button. The system then verifies the seats and processes the booking.

**Navigation Path:**

Movie Selection Screen > Seat Selection Screen > Booking Confirmation Screen



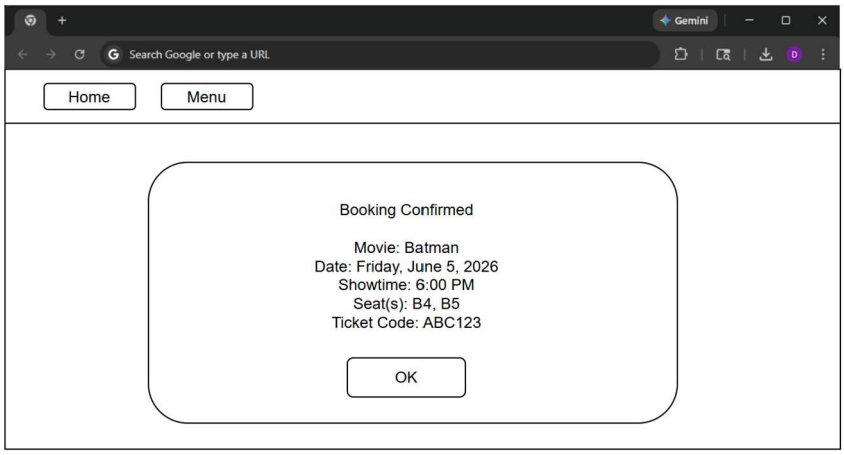

## Use Case 4: Display Booking Confirmation

**Preliminary Design**

After confirming the booking, the system displays a confirmation screen showing the movie title, date, showtime, selected seats, and ticket code. This screen confirms that the booking was successful and provides the user with ticket information.

**Navigation Path**

Seat Selection Screen > Booking Confirmation Screen



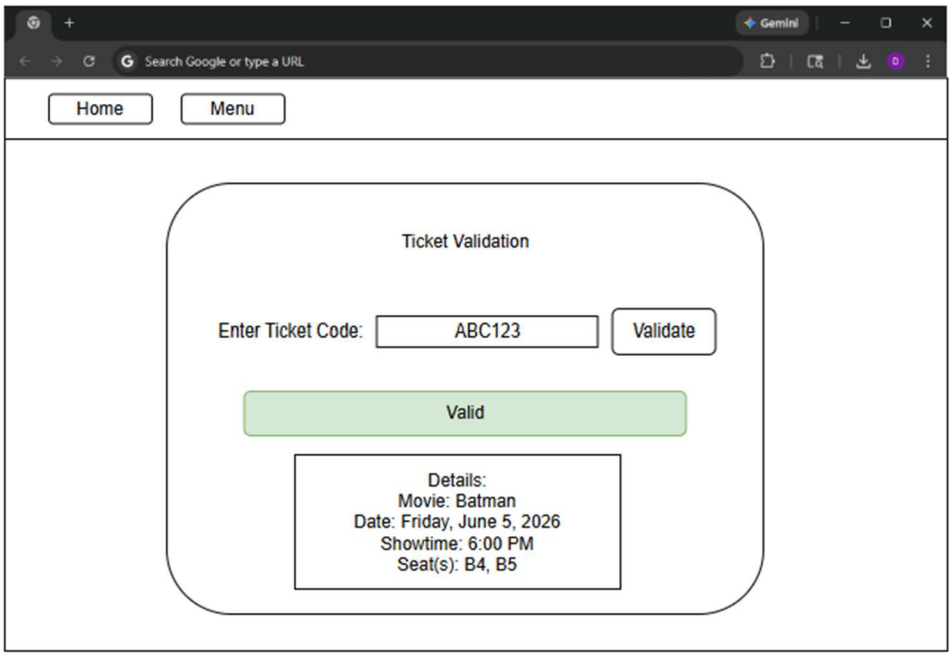
Use Case 5: Validate Ticket

**Preliminary Design:**

The ticket validation screen allows theater staff to scan or enter a ticket code. The system retrieves the booking record and checks whether the ticket is valid and unused. If valid, the system marks the ticket as used and allows entry. If invalid, the system displays an error message.

**Navigation Path:**

Staff Login > Ticket Validation Screen > Validation Result




## User Effort Estimation:

| Usage Scenario | Navigation | Clicks | Keystrokes |
|---|---|---|---|
| User Login | overview, enter email/password, login | 3 | <40 |
| View Movies and Showtimes | overview, select date, select showtime | 5 | 0 |
| Select Seats and Confirm | overview, select seat, confirm | 4 | 0 |
| Display Booking Confirmation | overview, okay | 1 | 0 |
| Validate Ticket | overview, enter ticket number, validate | 2 | <10 |

# Project Plan

| Week | Task | Status |
|---|---|---|
| 1-2 | Set up development environment, design database schema, connect front end to back end | Completed |
| 3-4 | Implement login and registration features | Completed |
| 5-7 | Develop manager features for movies, showtimes, and seating layouts | Completed |
| 8 | Midterm testing and demo recording | Completed |
| 9-11 | Develop customer booking features and seat reservation | Pending |
| 12-14 | Write test cases, fix bugs, improve usability | Pending |
| 15 | Final testing, documentation, and demo | Pending |

At the midpoint of the semester, the system design and documentation have been completed, including the system requirements, UML diagrams, activity diagrams, and user interface specifications. The project is currently transitioning from the design phase into the implementation phase. Initial development work has begun on setting up the development environment and planning the database structure that will store movies, showtimes, seats, and booking records.

For the remainder of the semester, the focus will be on implementing the core functionality of the booking system, including seat reservation logic, ticket generation, and ticket validation. Additional work will include improving the user interface and testing the system to ensure that bookings are processed correctly and seat availability updates reliably.

# References

Tilley, S., & Rosenblatt, H. J. (2017). Systems analysis and design (11th ed.). Cengage Learning.